Task 5: Emotion Classification Using Machine Learning

This laboratory exercise aims to investigate emotion classification in text using machine learning techniques.

1. **Text dataset preparation**

Use the metadata.csv file from the Lithuanian Emotional Speech Corpus (EMO) to create a text classification dataset.

- Extract the text samples and their corresponding emotion labels.
- Identify and exclude all texts associated with multiple emotion labels.
- Remove duplicate texts, ensuring that each unique text appears only once in the dataset.
- Save the resulting dataset in CSV format, containing the text samples and their emotion labels, for further analysis and classification.

2. **Text Analysis**

Analyze the linguistic characteristics of the prepared text dataset and investigate differences across the five emotion classes.

- Text length: Visualize and compare the distributions of text lengths (in words) across the emotion classes.
- Vocabulary analysis: Apply lemmatization and stop-word removal. Identify and compare the most frequent words in each emotion class. Generate a word cloud for each emotion class to visualize its most frequent words.
- Part-of-speech (POS) analysis: Analyze and compare the distributions of POS tags across the emotion classes.

3. **Text Representation (Feature Extraction)**

- Split the dataset into training and test sets using stratified sampling.
- Select and implement three text representation methods, choosing one from each category:
  1. Frequency-based representations: Bag-of-Words (BoW), TF-IDF, or other method.
  2. Static word embeddings: Word2Vec, GloVe, FastText, or another model.
  3. Contextual embeddings: BERT, Sentence-BERT, or other pretrained transformer model.
- Convert each text into a fixed-length feature vector suitable for classification. Ensure that any data-dependent feature extraction is fitted using only the training set.

4. **Emotion Classification**

Train a Logistic Regression classifier to predict the five emotion classes using each of the three selected text representations.

- Use the same training and test sets for all three representations.
- Train a separate classifier for each representation.
- Evaluate each classifier using accuracy, precision, recall, and F1-score.
- Generate and compare confusion matrices for all three models.

In the report:

- ✓ Report the number of texts associated with multiple emotion labels and excluded from the dataset. Report the number of text samples before and after duplicate removal. Present the distribution of texts across the five emotion classes.
- ✓ Summarize the main findings from the text analysis part, highlighting notable similarities and differences between the emotion classes.
- ✓ Describe and justify the selected text representation methods. Compare the resulting representations in terms of dimensionality, sparsity, and their ability to capture semantic information.

- ✓ Present the classification results in a comparative table. Discuss which text representation performs best and analyze the most common classification errors.

Report requirements:

The Python code should be included in the appendix.
Scope for report: **4-6 pages** (excluding the appendix and title page)

The report may be prepared using MS Word, LaTeX, or another document preparation environment of your choice. The final report must be submitted in **PDF** format. The report should follow the general formatting requirements applied to academic text. In particular, all figures and tables must be numbered and have informative captions; all figures and tables must be referred to and discussed in the text; etc.